

Quick Start Tests in CQ (SCF) (Energy Only)

1. Choose your test system (i.e. water)
2. Make input files
 1. Serial nsmp=1
 2. Parallel nsmp=2
3. Run test system
4. Place serial input file in:
 1. chronusq_dev/tests/scf/serial/your_method/
5. Place parallel input file in:
 1. chronusq_dev/tests/scf/parallel/your_method/
6. Do you need more than 1 bin file?
 1. If your energy should be the same for all tests then just keep 1 bin file
7. Place bin file (with ".ref" appended) in
 1. chronusq_dev/tests/scf/reference/
8. Update chronusq_dev/tests/scf/CMakeLists.txt

chronusq_dev/tests/scf/your_method.cxx

```
# Generate directories
file(MAKE_DIRECTORY ${SCF_TEST_BINARY_ROOT}/serial/your_method)
...
file(MAKE_DIRECTORY ${SCF_TEST_BINARY_ROOT}/parallel/your_method)
...
# Add tests
add_cq_test( YOUR_METHOD_SCF      scftest "YOUR_METHOD.*"      )
...
if( CQ_ENABLE_MPI )
  add_cq_mpi_test( YOUR_METHOD_SCF_MPI      2      scftest "YOUR_METHOD.*"      )
```

```
#include "scf.hpp"

// TEST DESCRIPTION test (details)
TEST( YOUR_METHOD, YOUR_INPUTFILE_NAME ) {

    CQSCFTTEST( "scf/serial/your_method/your_inputfile_name", "your_inputfile_name.bin.ref", 1e-6 );

};

// TEST DESCRIPTION test (slightly different)
TEST( YOUR_METHOD, YOUR_DIFFERENT_INPUTFILE_NAME ) {

    CQSCFTTEST( "scf/serial/your_method/your_different_inputfile_name", "your_inputfile_name.bin.ref", 1e-6 );

};

#ifdef _CQ_DO_PARTESTS

// TEST DESCRIPTION test (details)
TEST( YOUR_METHOD, YOUR_INPUTFILE_NAME ) {

    CQSCFTTEST( "scf/parallel/your_method/your_inputfile_name", "your_inputfile_name.bin.ref", 1e-6 );

};

// TEST DESCRIPTION test (slightly different)
TEST( YOUR_METHOD, YOUR_DIFFERENT_INPUTFILE_NAME ) {

    CQSCFTTEST( "scf/parallel/your_method/your_different_inputfile_name", "your_inputfile_name.bin.ref", 1e-6 );

};

#endif
```